

DHYAN THAKKAR

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Education

University of Minnesota - Twin Cities

MS in Robotics

Minneapolis, MN

August 2024 – December 2026

Institute of Technology, Nirma University

Bachelor's in Electronics and Communications

Ahmedabad, India

September 2020 – May 2024

Work Experience

Robotics Gallery, Science City

Robotics Intern

January 2024 – June 2024

Ahmedabad, India

- Developed realtime Visual-Inertial Odometry (VIO) on multicopters, reducing vertical path deviation by 5%
- Implemented LLM-based navigation using SLAM, leveraging 4D LiDAR and camera-based sensing on Unitree GO2
- Designed GSLAM-based AMR systems for transporting 200lb+ payloads in warehouse environments
- Maintained and demonstrated advanced robotic systems (AMRs, quadrupedal robots) for public educational exhibits

Electronics and Communications Department, Nirma University

Inhouse Intern

May 2023 – July 2023

Ahmedabad, India

- Deployed YOLOv8 with 30 FPS inference achieving 89% accuracy for vehicle classification and 67% precision in speed determination
- Trained custom dataset of 33,000+ images for real-time detection pipeline

Nirma University

Research Assistant

September 2021 – December 2022

Ahmedabad, India

- Streamlined remote development and containerization on multiple HPC systems and clusters
- Designed Bi-LSTM model to forecast cloud compute usage, improving resource scaling efficiency

Personal Projects

Real-time Human-to-Robot Motion Retargeting | *Isaac Sim, Mujoco, ROS*

October 2025

- Engineered low-latency pipeline utilizing Gravity-Constrained Mesh Reconstruction for 3D human sequences from monocular RGB
- Implemented Gaussian Mixture Regression (GMR) and TWIST for motion mapping, achieving stable 30 FPS on 6GB VRAM
- Benchmarked against AMASS dataset for high-fidelity motion mimicking on humanoid platforms in physics-based simulations

Online System Identification for Quadrupedal Pushing/Pulling | *ANYmal, Unitree Go1*

December 2025

- Developed adaptive framework for quadrupeds to perform online estimation of object properties (Mass, CoM, Moments of Inertia)
- Achieved 89% success rate on goal-oriented manipulation tasks with mean angular error ≤ 10 degrees
- Optimized full-body control strategies for dynamic interactions with unknown payloads in real-time environments

Anthropomorphic Trajectory Optimization | *Python, Kinematic Constraints*

January 2026

- Investigated robot arm trajectory parameterization aligned with natural human kinematic constraints and velocity profiles
- Developed velocity metrics for reaching tasks to improve predictability and safety in shared human-robot environments

Visual Tokens based Navigation | *Python, VLMs, Hugging Face*

April 2025

- Engineered navigation system using QWEN2.5VL 8B with scene graphs in 4-person team
- Fine-tuned Qwen2.5 and InternVL3 for visual scene grounding, achieving 78.64% on ObjectNav and 62.35% on OVMM
- Boosted landmark pose estimation by 27% using camera intrinsics as tokens to hybrid VLM model

Object Centric RL Based UAV Control | *Python, ISAAC Sim, PX4, OpenCV*

November 2024

- Reduced VIO jumps by 80% using RL-based UAV control with RGB-D, EKF pose estimation, and segmentation-based detection
- Trained PPO agent for hovering and orbiting targeted objects inside ISAAC Sim/Lab

LLM-based Navigation with LiDAR on Unitree Go2 | *Python, OpenCV, ChatGPT, SDK2*

April 2024

- Deployed conversational agent on Unitree Go2 quadruped with 15+ retrievable goal landmark tokens
- Navigated 1700+ sq.ft. dynamic indoor environment with real-time path planning and localization
- Achieved sub-782ms latency on custom audio pipeline for edge-deployed finetuned LLM

Technical Skills

Languages: Python, C/C++, CUDA, Shell, MATLAB, Docker

Simulation & Physics: NVIDIA Isaac Sim/Lab, Mujoco, Gazebo, Habitat Sim, AWS Robomaker

Robotics Frameworks: ROS (Noetic, Humble, Jazzy), Nav2 Stack, Unitree SDK2, PX4, MAVROS

Machine Learning: PyTorch, Hugging Face, RLLib, Deepspeed, MMCV, YOLOv8, Bi-LSTM

Hardware & Systems: Jetson Nano, Raspberry Pi, Unitree Go1/Go2, ANYmal, Agate HPC Cluster

Computer Vision: RGB-D Sensing, EKF Pose Estimation, vSLAM, G-SLAM, Intel RealSense

Datasets: AMASS, nuScenes3D, Waymo Open, KITTI, Habitat-Matterport 3D, HSSD, RxR